Exact Inference Is Not Enough

Opponent models and endgames in six-player Literature — KRAKEN v1.1

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Literature (also Fish, or Canadian Fish) is a six-player, two-team, imperfect-information card game. Players interrogate opponents for named cards and score by declaring exactly how a six-card half-suit is split among their own team. Every action is public, so the whole hidden state is the deal — and there is no legal communication channel, which puts the game outside both the two-player zero-sum machinery that solved poker and the fully cooperative machinery built for Hanabi. This report is four development cycles on an engine for it, written with the arms that failed left in.

I am writing because I would value criticism of the method, and because one episode in it seems worth more than the engine is.

The result I retracted, and how

An earlier version of this work reported that KRAKEN beat an independently developed C++ engine by $+2.347$ sets per game over 10,000 audited games. **That figure is withdrawn.** It measured my bridge, not my engine.

The bridge was stateless: it replayed the entire public log into a freshly reset copy of their agent at every decision, where their own arbiter keeps one agent per seat alive for the whole deal. Those are the same thing only if their agent is a pure function of (reset state, event sequence). It is not.

What made this findable was not suspicion of the number. It was a control. To check that their engine had not simply got weaker, I ran their whole released ladder, v0.2 through v0.7, and the ladder came out *backwards*: in my arbiter their later releases scored worse, and their declaration error climbed monotonically with their own version number (Spearman -1.000 against $+0.771$ the other way). A project's successive releases getting steadily worse at the one thing the game scores is not a shape that improving strength produces. So the ladder was the instrument, not the finding.

Localising it took a decision-level parity test — the same game put to both transports, decision by decision:

their version	decisions	divergent	rate
v0.2 (scripted)	323	**0**	0.00%
v0.3 (scripted)	338	**0**	0.00%
v0.4	415	81	19.52%
v0.5	336	137	40.77%
v0.6	292	36	12.33%
v0.7	296	84	28.38%

Exactly zero on both of their scripted baselines, and large on every release from v0.4 — the version their fitted belief arrives at. A clean boundary, and the same rung at which the two arbiters' ladders stop agreeing.

Then it was priced, under a paired design that holds the deal, the seats, the agent seeds and my own engine fixed and changes only whether their process survives between decisions. **The bridge was worth** $+3.18$ $[+3.01, +3.34]$ **sets a game to me** — more than the entire margin it produced. Their declaration errors fall roughly tenfold, from 0.880 a game to 0.092, once their engine is allowed to keep the state its own arbiter lets it keep.

Corrected, KRAKEN **loses** to that engine by -0.70. Measured independently inside *their* arbiter, through a bot-package protocol they published, by -0.62. Two routes that share no host, no rules implementation, no deal generator and no line of code agree to within 0.08 — where the published number was out by three and had the sign wrong.

The general lesson is one the report had already written down in an appendix and then not applied to its own headline: *an absolute measured through a bridge is a statement about the bridge as well as about the engines.* Paired contrasts that hold the bridge fixed on both sides were unaffected, and are what the rest of the report rests on.

The central result is also negative

Hidden-state inference here reduces to counting assignments in a degree-constrained bipartite system. That is #*P*-complete in general, but real positions carry only 4.1 distinct candidate sets across 25.1 unlocated cards, so a dynamic program returns the exact partition function, marginals, uniform samples and joint probabilities — validated against exhaustive enumeration.

Making the posterior exactly correct **changed nothing in play** (-0.008 sets per duplicate deal-pair over 250 pairs) and made it a *worse* predictor of where cards actually sit than the biased sampler it replaced (1.3618 against 1.3408 mean negative log-likelihood). A uniform posterior is a correct theorem about a false hypothesis: it assumes choices carry no information beyond legality. The old sampler's bias had accidentally encoded an opponent model, and removing the bias removed the model. Replacing that accident with an explicit one-parameter choice model is worth $+1.92$ $[+1.48, +2.36]$ sets per pair — the largest single gain in the study.

Two of the results that follow. Residual errors are not failures to *find* cards: 95.3% are *allocation* errors, where the team held all six of a half-suit and split them wrongly. Handing a seat the true deal one side at a time — a ceiling obtained by cheating, never a strength figure — prices teammates' cards at $+3.41$ against opponents' $+1.31$, and I had predicted the reverse.

How the claims are held to account

Forty-four pre-registrations, written before the runs, each naming its prediction and its ship bar; Appendix A tabulates every one on the same footing, *including* those whose predictions failed, and a test checks that table against the directory both ways. Nothing ships below ± 0.15 sets per game. Every number in the report is pinned by a script to the artifact that produced it, and the script is verified by perturbing an artifact and confirming it fails. Cross-engine work is pinned to an upstream commit and re-pinned only on evidence: 6,329 decisions replayed through binaries built at both ends, zero mismatches.

None of that caught the bridge. What caught it was running a control that had no reason to be interesting.

What is open

Which piece of their state the bridge discards is only partly separated, and what I have went against the convenient reading. Their determinized search is the largest term I can name, and the one a fix could reach — their reset re-seeds it. But sorting the divergences into disagreements about *which* ask to make and disagreements about *whether to declare at all* puts everything I can identify in the first, while the second — the band that costs sets, since a declaration taken a turn early is a wrong one — does not move at all. Two mechanisms I had believed in did not survive their source, and 13.92% of decisions still differ with everything I found switched off. So the bridge cannot be repaired by re-seeding, and I do not yet know what the remainder is. The $+3.18$ price rests on 2,000 pairings: an order of magnitude past its interval, but I would want more games before quoting it alone.

I would be glad of any reaction, and particularly of the sharpest version of "that control only worked by luck." I think it did, and I do not yet know what to put in its place.

Report, code, pre-registrations and result artifacts: `https://github.com/kv1514/fish-researchp12`